Exercise 1: Meet Your Data – SOLUTIONS

Machine Learning with R in the Life Sciences

Setup

```r
library(palmerpenguins)
library(tidyverse)
data("penguins")
```

1.1

```r
dim(penguins)
```

```
## [1] 344   8
```

```r
head(penguins)
```

```
## # A tibble: 6 x 8
##   species island    bill_length_mm bill_depth_mm flipper_length_mm body_mass_g
##   <fct>   <fct>              <dbl>         <dbl>             <int>       <int>
## 1 Adelie  Torgersen           39.1          18.7               181        3750
## 2 Adelie  Torgersen           39.5          17.4               186        3800
## 3 Adelie  Torgersen           40.3          18                 195        3250
## 4 Adelie  Torgersen           NA            NA                  NA          NA
## 5 Adelie  Torgersen           36.7          19.3               193        3450
## 6 Adelie  Torgersen           39.3          20.6               190        3650
## # i 2 more variables: sex <fct>, year <int>
```

```r
summary(penguins)
```

```
##       species          island    bill_length_mm  bill_depth_mm
##  Adelie   :152   Biscoe   :168   Min.   :32.10   Min.   :13.10
##  Chinstrap: 68   Dream    :124   1st Qu.:39.23   1st Qu.:15.60
##  Gentoo   :124   Torgersen: 52   Median :44.45   Median :17.30
##                                   Mean   :43.92   Mean   :17.15
##                                   3rd Qu.:48.50   3rd Qu.:18.70
##                                   Max.   :59.60   Max.   :21.50
##                                   NA's   :2       NA's   :2
##  flipper_length_mm  body_mass_g       sex           year
##  Min.   :172.0     Min.   :2700   female:165   Min.   :2007
##  1st Qu.:190.0     1st Qu.:3550   male  :168   1st Qu.:2007
##  Median :197.0     Median :4050   NA's  : 11   Median :2008
```

```
##  Mean   :200.9     Mean   :4202                Mean   :2008
##  3rd Qu.:213.0     3rd Qu.:4750                3rd Qu.:2009
##  Max.   :231.0     Max.   :6300                Max.   :2009
##  NA's   :2         NA's   :2
```

1.2

```r
str(penguins)
```

```
## tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
##  $ species          : Factor w/ 3 levels "Adelie","Chinstrap",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ island           : Factor w/ 3 levels "Biscoe","Dream",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ bill_length_mm   : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
##  $ bill_depth_mm    : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
##  $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
##  $ body_mass_g      : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
##  $ sex              : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
##  $ year             : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

Q: Which columns are numeric? Which are categorical? Which column could be a target variable (something you would want to predict)?

Numeric columns: `bill_length_mm`, `bill_depth_mm`, `flipper_length_mm`, `body_mass_g`, and `year` (technically numeric but functions as a categorical grouping variable here — there are only 3 unique years).

Categorical columns: `species`, `island`, and `sex`.

You can verify this with `str(penguins)` — numeric columns show as `num` or `int`, categorical ones as `fct` (factor).

Target variable: The most natural target is `species` — predicting which species a penguin belongs to based on its body measurements. This is a classification task with three classes.

Other valid targets depending on the research question:

- `sex` — predicting sex from morphology (binary classification)
- `body_mass_g` — predicting body mass from the other measurements (regression)
- `island` — predicting island of origin (classification, though this is partly confounded with species)

The choice of target depends on the scientific question. In ML terms, the target is always the column you want to predict, and everything else becomes a candidate feature.

1.3

```r
penguins_clean <- penguins |>
  drop_na()
dim(penguins_clean)
```

```
## [1] 333   8
```

1.4

```r
penguins_clean |>
  count(species)
```

```
## # A tibble: 3 x 2
##   species       n
##   <fct>     <int>
## 1 Adelie      146
## 2 Chinstrap    68
## 3 Gentoo      119
```

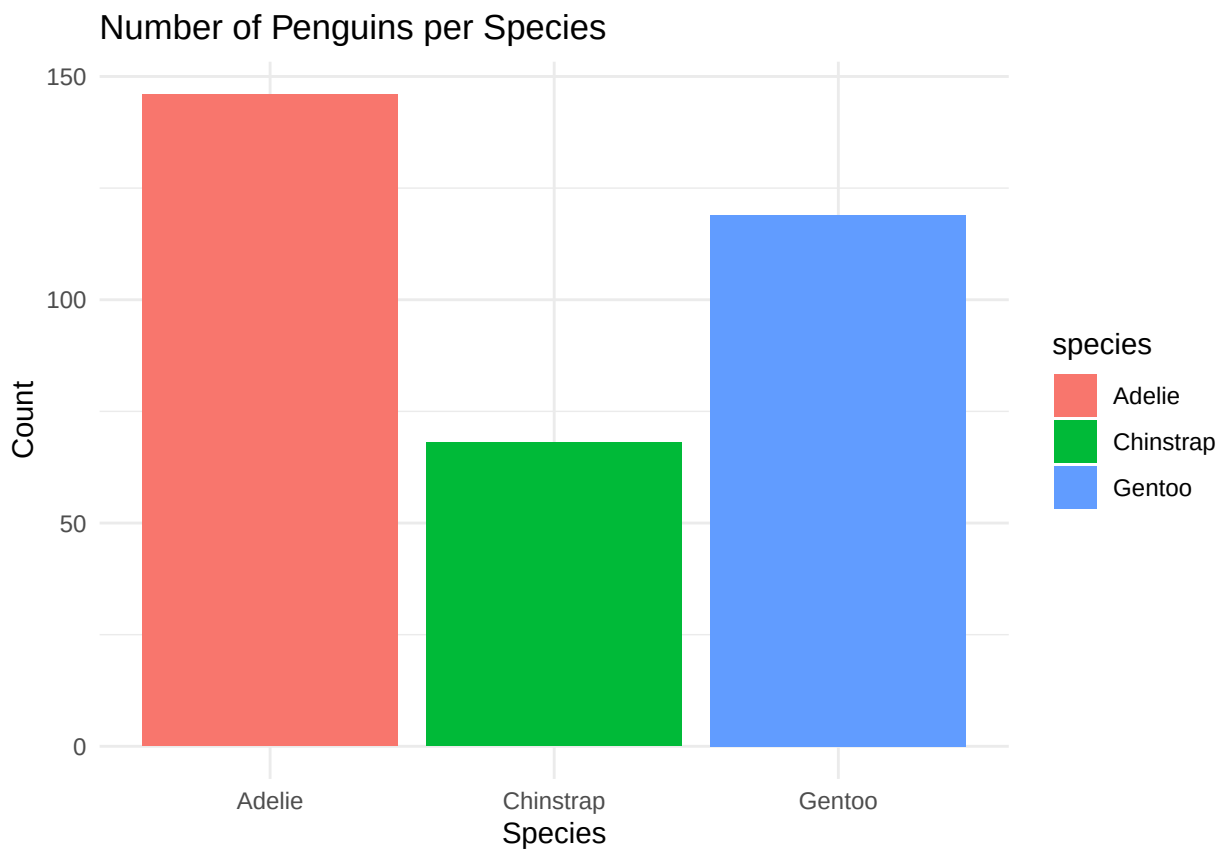
1.5

```r
penguins_clean |>
  group_by(species) |>
  summarize(mean_mass = mean(body_mass_g))
```

```
## # A tibble: 3 x 2
##   species   mean_mass
##   <fct>         <dbl>
## 1 Adelie        3706.
## 2 Chinstrap     3733.
## 3 Gentoo        5092.
```

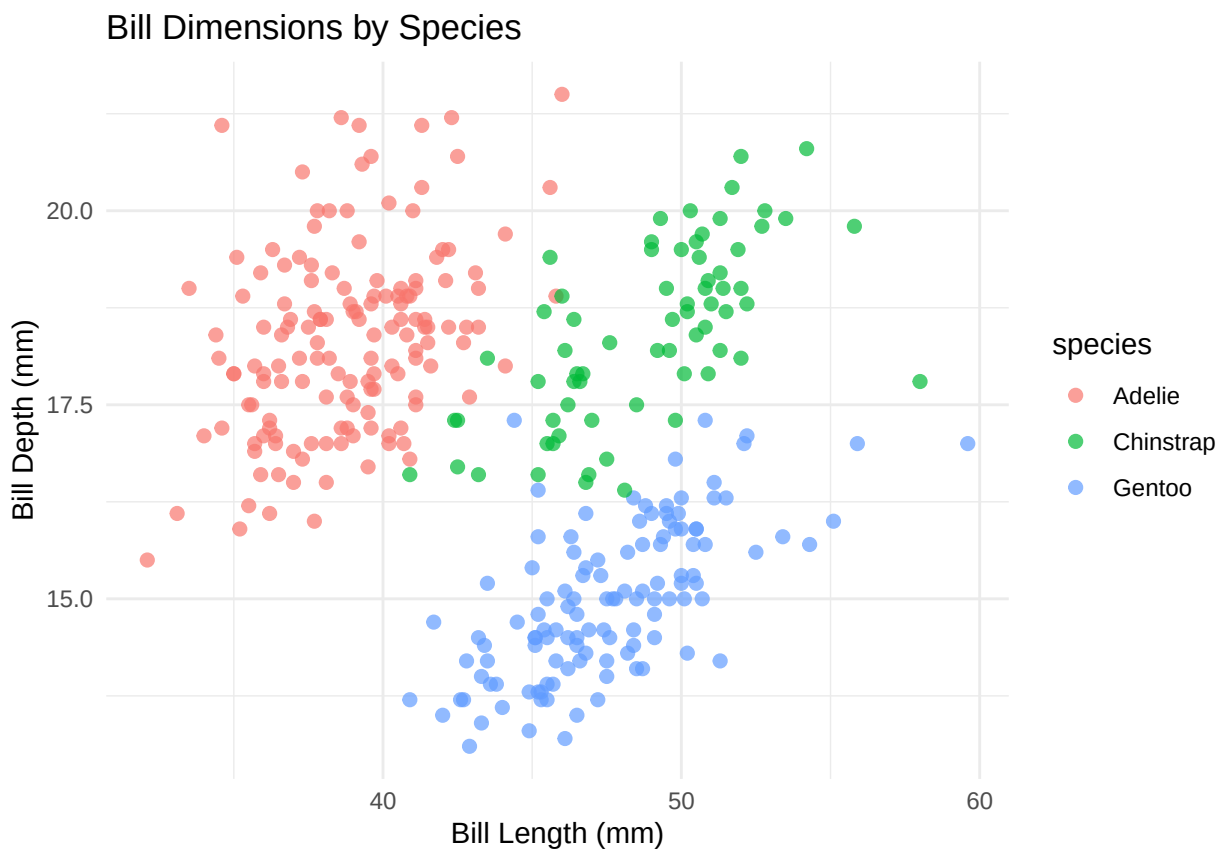
1.6

```r
ggplot(penguins_clean, aes(x = species, fill = species)) +
  geom_bar() +
  theme_minimal() +
  labs(title = "Number of Penguins per Species", x = "Species", y = "Count")
```



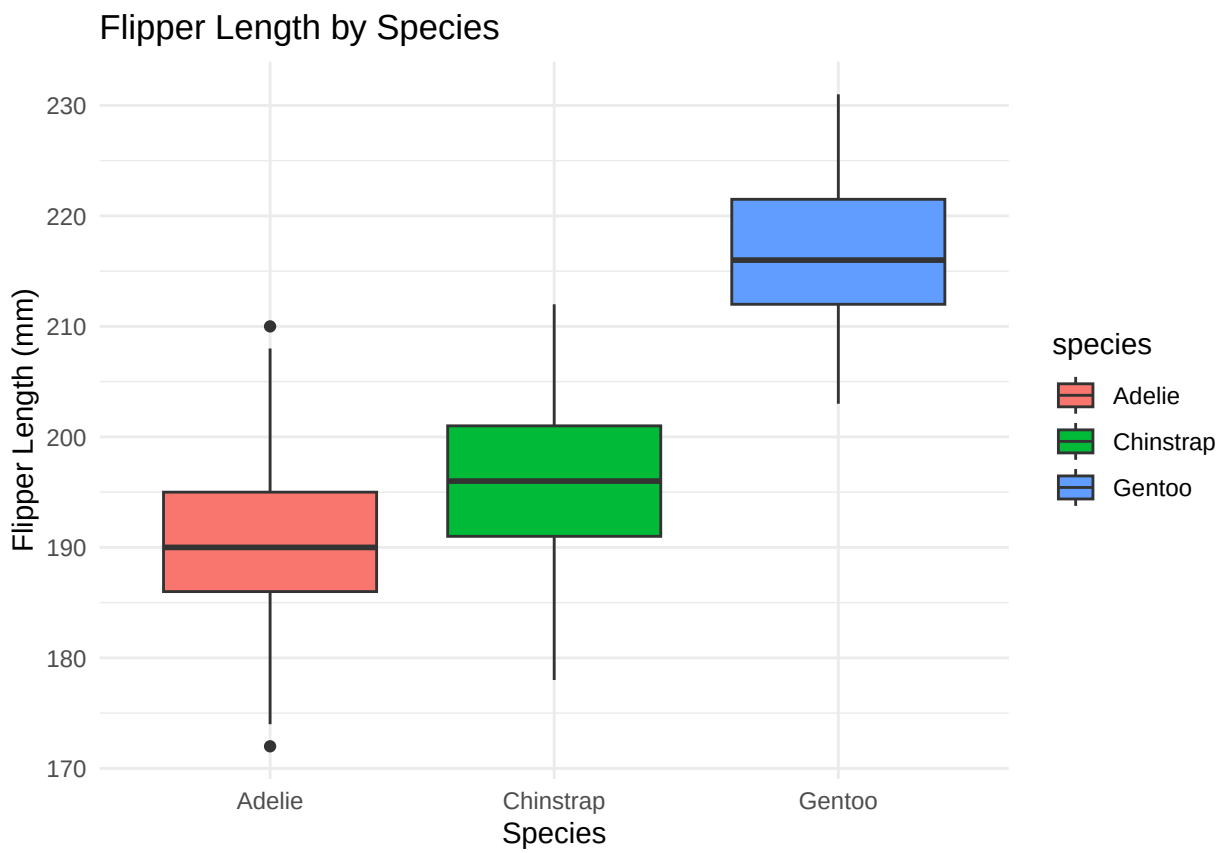
1.7

```
ggplot(penguins_clean, aes(x = bill_length_mm, y = bill_depth_mm, color = species)) +
  geom_point(alpha = 0.7, size = 2) +
  theme_minimal() +
  labs(title = "Bill Dimensions by Species", x = "Bill Length (mm)", y = "Bill Depth (mm)")
```



1.8

```
ggplot(penguins_clean, aes(x = species, y = flipper_length_mm, fill = species)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Flipper Length by Species", x = "Species", y = "Flipper Length (mm)")
```



Discussion Questions — Solutions

Q1: If we wanted to predict the species of a penguin from its measurements, which columns would be the features (inputs) and which would be the target (output)?

The **target** (output) is the `species` column — that is what we want to predict. The **features** (inputs) are the four numeric measurement columns: `bill_length_mm`, `bill_depth_mm`, `flipper_length_mm`, and `body_mass_g`.

The columns `island`, `sex`, and `year` could also serve as features, but they are categorical and would require encoding. For a first ML exercise we focus on the numeric measurements because they directly describe the penguin's morphology and are biologically meaningful for distinguishing species.

Q2: Is this a classification or a regression problem? Why?

This is a **classification** problem because the target variable `species` is categorical (Adelie, Chinstrap, Gentoo) — we are predicting a discrete class label, not a continuous number. If we were instead predicting `body_mass_g` from the other measurements, *that* would be a regression problem.

Q3: Looking at your scatter plot from 1.7: do you think a machine learning model could separate the three species? Which species pair might be hardest to distinguish?

Yes. The scatter plot of `bill_length_mm` vs. `bill_depth_mm` shows that the three species form relatively distinct clusters, so an ML model should be able to separate them reasonably well.

Adelie and Chinstrap are the hardest pair to distinguish — they overlap substantially in bill depth and have adjacent ranges in bill length. Gentoo penguins form a clearly separated group (lower bill depth, higher bill length), making them the easiest to identify.

This illustrates why we often need multiple features: adding flipper length and body mass would help separate Adelie from Chinstrap further.

Q4: We have 333 samples. Is this a lot or a little for ML? What if we had 33,000 genes instead of 4 numeric features?

For 4 numeric features, 333 samples is quite reasonable — a common rule of thumb is at least 10–20 observations per feature, and we far exceed that.

However, if we had **33,000 genes** instead of 4 features, the situation reverses dramatically: we would have far more features than samples (p » n). This is the **curse of dimensionality** — the model has too many degrees of freedom relative to the data and will overfit to noise. In that high-dimensional setting we would need:

- Feature selection (e.g., variance filtering, selecting only DE genes)
- Dimensionality reduction (PCA)
- Regularized models (Lasso, Elastic Net)

This p » n scenario is extremely common in transcriptomics, proteomics, and other omics data.